



Amazon Elastic Block Store (EBS)

Lecture Slides

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Good luck on your exam!

Introduction to EBS

What is IOPS?

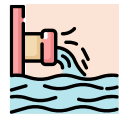
IOPS stands for Input/Output Per Second. It is the speed at which **non-contiguous reads and writes** can be performed on a storage medium. High I/O = lots of small, fast reads and writes

What is Throughput?

The data transfer rate to and from the storage medium in megabytes per second.

What is Bandwidth?

Bandwidth is the measurement of the total possible speed of data movement along the network



Think of **Bandwidth** = **Pipe** and **Throughput** = **Water**

Introduction to EBS



Elastic Block Store (EBS) is a highly available and durable solution for attaching persistent block storage volumes to an EC2 instance.

Volumes are automatically replicated within their Availability Zone (AZ) to protect from component failure.

The types of volumes you can deploy:

- **General Purpose SSD (gp2)** — for general usage without specific requirements
- **General Purpose SSD (gp3)** — up to 20% lower cost per GB than gp2
- **Provisioned IOPS SSD (io1)** — when you require really fast input & output
- **Provisioned IOPS SSD (io2)** — more durable than io1
- **io2 Block Express** — higher throughput and IOPS and support larger storage capacity
- **Cold HDD (sc1)** — Lowest cost HDD volume for infrequently access workloads
- **Throughput Optimized HDD (st1)** — magnetic drive optimized for quick throughput
- **Magnetic (standard)** — previous generation HDD

All io2 volumes created after **November 21, 2023** are io2 Block Express volumes.
io2 volumes created before **November 21, 2023** can be converted to io2 Block Express volumes

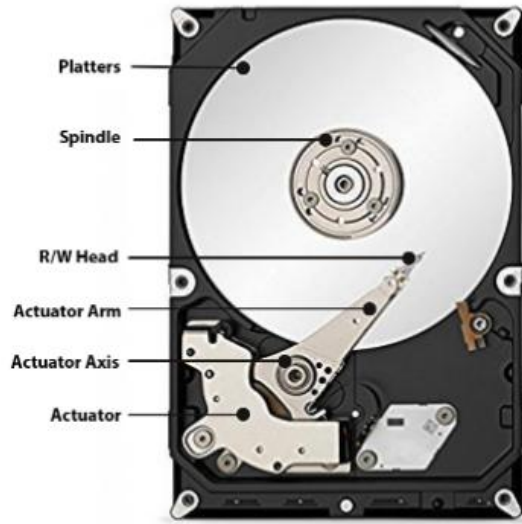
EBS – Volume Type Usage

	General Purpose SSD volumes		Provisioned IOPS SSD volumes	
Volume Type	gp2	gp3	io1	io2 Block Express
Use Case	Most general workloads • Transactional workloads • Virtual desktops • Medium-sized, single-instance databases • Low-latency interactive applications • Boot volumes • Development and test environments		• Workloads that require sustained IOPS performance or more than 16,000 IOPS • I/O-intensive database workloads	• Sub-millisecond latency • Sustained IOPS performance • More than 64,000 IOPS or 1,000 MiB/s of throughput
Durability	99.8% - 99.9%		99.8% - 99.9%	99.999%
Volume Size	1 GiB – 16 TiB		4 GiB– 16 TiB	4 GiB– 64 TiB
Max IOPS	16,000 (16 KiB I/O)		64,000 (16 KiB I/O)	256,000 (16 KiB I/O)
Max Throughput	250 MiB/s	1,000 MiB/s	1,000 MiB/s	4,000 MiB/s
EBS Multi-attach	Not Supported		Supported	
NVMe Reserv.	Not Supported			Supported
Boot Volume	Supported			

EBS – Volume Type Usage

	Throughput Optimized HDD	Cold HDD	Magnetic
Volume Type	st1	sc1	standard
Use Case	• Big data • Data warehouses • Log processing	• Throughput-oriented storage for data that is infrequently accessed • Where lowest storage cost is important	• Workloads where data is infrequently accessed
Durability	99.8% - 99.9%		N/A
Volume Size	125 GiB - 16 TiB		1 GiB - 1 TiB
Max IOPS	500 (1 MiB I/O)	250 (1 MiB I/O)	40-200
Max Throughput	500 MiB/s	250 MiB/s	40-90 MiB/s
EBS Multi-attach	Not Supported		N/A
Boot Volume	Not Supported		Supported

Storage Volumes – HDD



Hard Disk Drive (HDD)

HDDs are traditional magnetic storage devices that use spinning platters and a moving read/write head (similar to a record player) to store and retrieve data

- **Better for Throughput:** Ideal for sequential data workloads, such as log processing and data backups
- **Physical Moving Part:** Mechanical components make them slower than SSDs and more prone to wear and tear

RPMs (Revolutions Per Minute)

RPM measures how fast the platters inside the drive spin. In AWS, RPM is abstracted but still impacts performance tiers

7,200 RPM Drives:

Standard for desktops and high-performance workloads. AWS **st1** volumes offer comparable performance.

5,400 RPM Drives:

Common in laptops and external hard drives. Used for archival storage. AWS **sc1** volumes align with this tier.

10,000+ RPM Drives:

Designed for enterprise environments with critical performance needs. AWS SSD-based options often outperform these.

Storage Volumes – HDD RAID

RAID (redundant array of independent disks)

RAID is a **data storage virtual technology** for magnetic disks (*not magnetic tape*) to improve fault tolerance.

RAID combines multiple physical volumes into one logical group, storing redundant data across disks

Since HDD has with mechanical parts and will result in wear HDD is more prone to failure than SSD.

RAID 0 (Striping)

- No redundancy; data is split across disks for high performance.
- Increases speed and capacity but offers no fault tolerance.
- Minimum of 2 disks required.

RAID 1 (Mirroring)

- Data is duplicated on two or more disks, offering high redundancy.
- If one disk fails, data is still accessible from another.
- Requires at least 2 disks.

RAID 5 (Striping with Parity)

- Combines striping and parity for both speed and data protection.
- Can withstand the failure of one drive without data loss.
- Requires at least 3 disks.

RAID 6 (Striping with Double Parity)

- Similar to RAID 5 but with double parity, allowing it to survive the failure of two disks.
- Requires at least 4 disks.

RAID 10 (1+0)

- A combination of RAID 1 and RAID 0, offering redundancy and increased performance.
- Requires a minimum of 4 disks.

Storage Volumes – SSD

Solid State Drive (SSD)

Uses integrated circuit (IC) assemblies as memory to store data persistently, typically using **flash memory**.

SSDs are more resistant to physical shock, run silently, and have quicker access time and lower latency.

- Very good frequently reads and writes (I/O)
- No physical moving parts
- **SATA SSDs:** Widely used, compatible with most computers, offering good performance but slower than NVMe due to SATA interface limitations.
- **NVMe SSDs:** Use the PCIe interface for higher performance. Ideal for intensive data tasks and gaming, available in M.2 or PCIe card form factors.
- **M.2 SSDs:** Compact, suitable for laptops and compact PCs, can use SATA or NVMe interfaces. They are installed directly on the motherboard.
- **U.2 SSDs:** Similar in performance to M.2 NVMe SSDs but designed for 2.5-inch drive bays, mainly used in enterprise and server environments.
- **Portable SSDs:** External drives for easy portability, connecting via USB or Thunderbolt. Offer SSD speed and durability on the go.
- **PCIe SSDs:** Add-on cards that provide high performance for older systems or specialized tasks, fitting into the motherboard's PCIe slots.

NAND Flash Memory

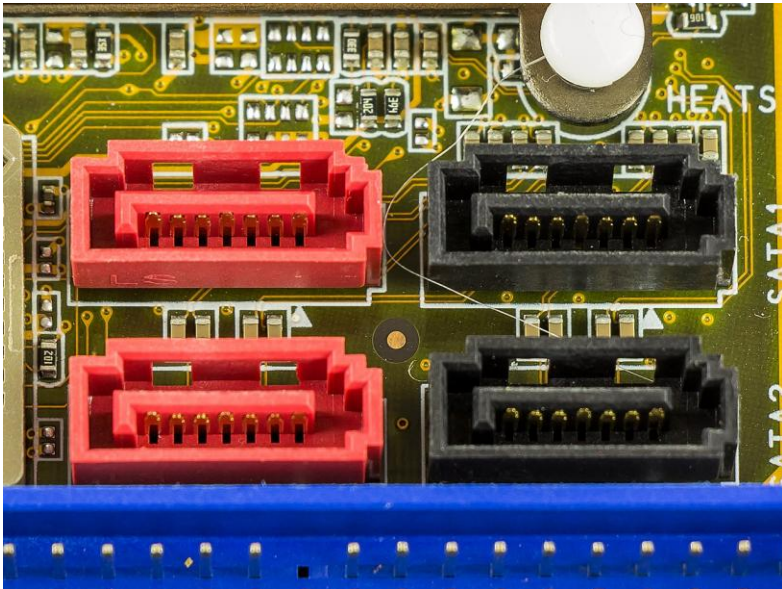


Storage Volumes - SSD

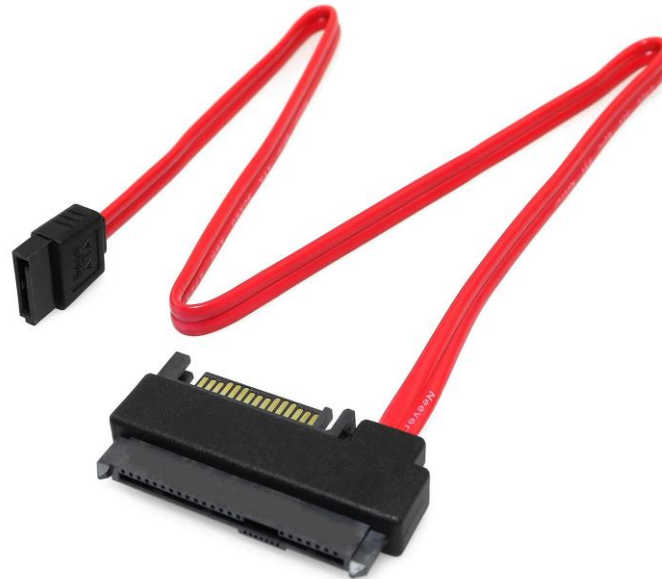
Serial Advanced Technology Attachment (SATA) SSDs:

These are the most common and are compatible with the Serial ATA interface, used by most desktop and laptop computers. They are relatively easy to install and offer good performance, though they are typically slower than their NVMe counterparts due to the limitations of the SATA interface.

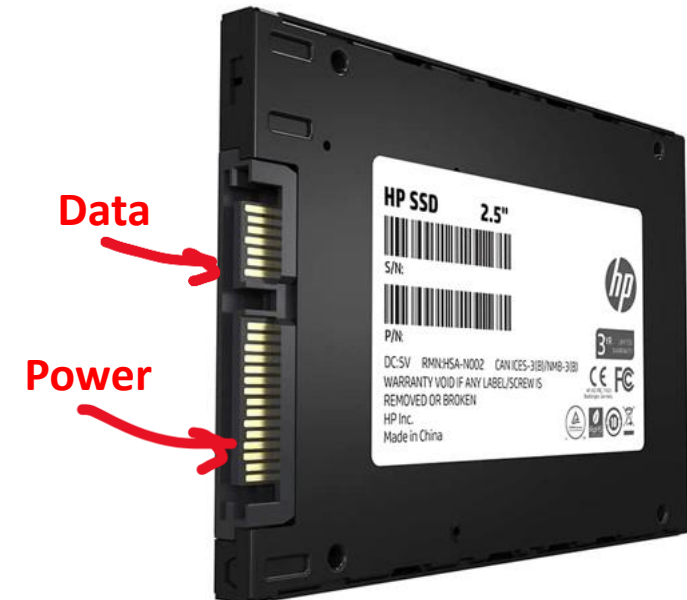
SATA Ports



SATA Cable



SATA SSD Volume



Storage Volumes – SSD

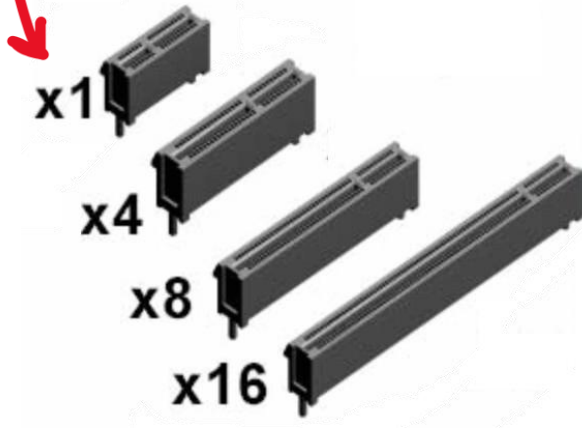
(Peripheral Component Interconnect Express) PCIe SSDs:

These SSDs are add-on cards that fit into the PCIe slots on a motherboard.

They can offer very high performance, especially in configurations that support NVMe.

They are a good option for users looking to upgrade older systems or for specialized high-performance tasks.

PCIe slots on a motherboard come in **different sizes**, referred to as “lanes”



PCIe Slot Generation	Released	Max Bandwidth Per Lane	Max Bandwidth x16 Slot
PCIe 1.0	2003	250 MB/s	4 GB/s
PCIe 1.1	2005	250 MB/s	4 GB/s
PCIe 2.0	2007	500 MB/s	8 GB/s
PCIe 3.0	2010	1 GB/s	16 GB/s
PCIe 4.0	2017	2 GB/s	32 GB/s
PCIe 5.0	2019	4 GB/s	64 GB/s
PCIe 6.0	2022	8 GB/s	128 GB/s

Storage Volumes – SSD

(Non-Volatile Memory Express) NVMe SSDs

NVMe SSDs use the PCIe interface, offering significantly higher performance compared to SATA SSDs. They are designed to take full advantage of the high speeds of flash-based storage technologies. NVMe drives are found in the form of M.2 SSDs but can also be added to a computer using a PCIe expansion slot.

M.2 SSDs

Compact, suitable for laptops and compact PCs, **can use SATA or NVMe interfaces**. They are installed directly on the motherboard.



U.2 SSDs

Similar in performance to M.2 NVMe SSDs but designed for 2.5-inch drive bays, mainly used in enterprise and server environments. They generally connect via a U.2 port



Storage Volumes – Magnetic Tape



Magnetic Tape

A large reel of magnetic tape. A tape drive is used to write data to the tape. Medium and large-sized data centers deployed both tape and disk formats. They normally come in the form of a cassette. Magnetic is very cheap to produce and can store a considerable amount of data.

- Durable for decades (good for at least up to 30 years)
- cheap to produce

A Tape drive that reads modern cassette



Modern cassettes containing magnetic tape

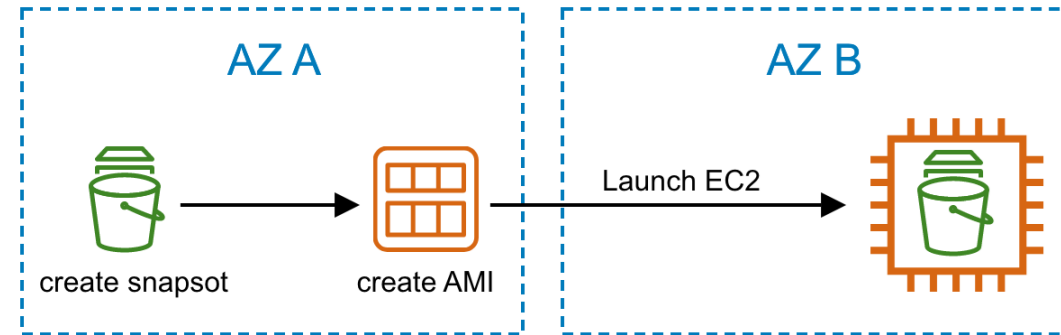


EBS – Moving Volumes

To move EBS volumes you need to make copies of those volumes as “snapshots”

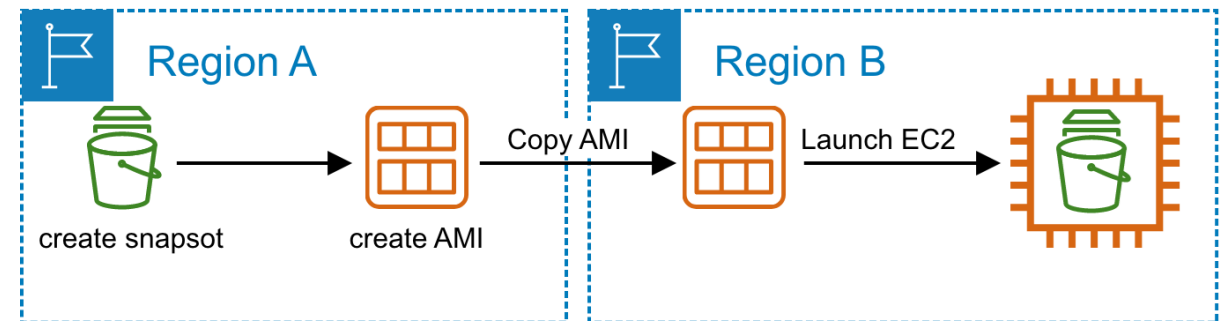
From one AZ to another

1. take a Snapshot of the volume
2. create an AMI from the Snapshot
3. launch a new EC2 instance in desired AZ



From one Region to another

1. take a Snapshot of the volume
2. create an AMI from the Snapshot
3. copy the AMI to another region
4. launch a new EC2 instance from the copied AMI.



EBS - Encrypted Root Volume

EBS Root Volumes **are not encrypted by default.**

You need to turn it on at the time of creating an EBS volume or at the time of creating your EC2 instance



Encryption [Info](#)

Use Amazon EBS encryption as an encryption option

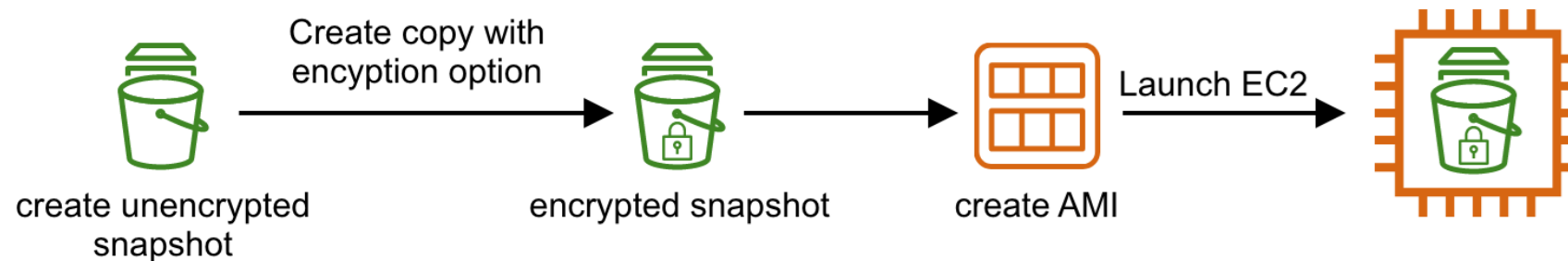
☒ Encrypt this volume

KMS key [Info](#)

Select a KMS key

If you want to encrypt an **existing volume**, you'll have to do the following:

- Take a Snapshot of the unencrypted volume
- Create a copy of that Snapshot and select the **Encryption option**
- Create a new AMI from the encrypted Snapshot
- Launch a new EC2 instance from the created AMI



EBS Volumes vs Instance Store Volumes

An EC2 instance will be backed (a root device) by an **EBS Volume** or **Instance Store Volume**

EBS Volumes

A **durable**, block-level storage device that you can attach to a single EC2 instance

EBS Volume can be created from an EBS Snapshot

- Can start and stop instances.
- Data will persist if you reboot your system

*Ideal for when you want data to persist.
In most use cases, you'll want EBS-backed volume*

Instance Store Volumes (Ephemeral)

A **temporary** storage type located on disks that are physically attached to a host machine.

An Instance Store Volume is created from a template stored on the host machine

- Cannot stop instances using Instance Store Volumes. Can only terminate.
- Data will be lost if the host fails or the instance is terminated

Ideal for temporary backup and storing an application's cache, logs, or other random data.

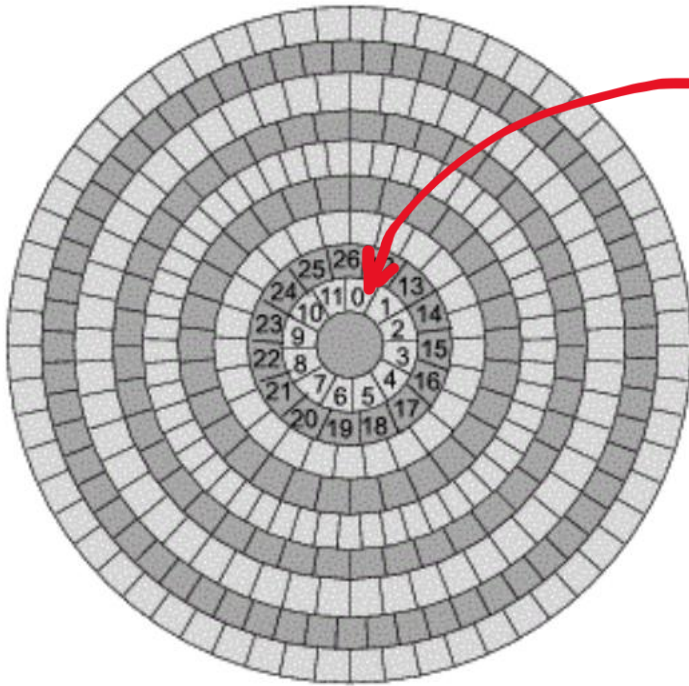
Partitioning Schemes

MBR (Master Boot Record) and GPT (GUID Partition Table) are two different methods of storing partitioning information on a disk. They are fundamentally different in how they manage this information and their capabilities.

	MBR (Master Boot Record)	GPT (GUID Partition Table)
Compatibility	Older operating systems and BIOS	Modern operating systems and UEFI firmware
Partition Limit	4 primary partitions or 3 primary and 1 extended with multiple logical partitions	Up to 128 partitions (no need for extended/logical partitions)
Disk Size Limit	2 TB	2 TB – 9.4 ZB (exabytes)
Partitioning Scheme	Partition Table is stored in the MBR itself	Uses a separate partition for the partition table, known as the EFI System Partition (ESP)
Boot Process	BIOS looks for boot loader in the first sector of the disk	UEFI reads the EFI System Partition (ESP) to start the boot loader
Unique Identifier	Does not support unique identifiers for partitions	Supports globally unique identifiers (GUIDs) for partitions

Logical Address Blocking

Logical block addressing (LBA) is a common scheme used for specifying the location of blocks of data stored on storage volumes.



blocks are located by an integer index, with the first block being LBA 0, the second LBA 1, and so on

Different block sizes for logical addressing in file systems and storage devices are used to balance efficiency, performance, and the management of storage space.

- Block size can be specified during formatting a storage volume.
- Block size has a direct impact on maximum volume size.
- The industry default size for logical data blocks is currently **4,096 bytes (4 KiB)**.

Block Size and Max Volume Size

- 4 KiB — 16 TiB (industry default size)
- 8 KiB — 32 TiB
- 16 KiB — 64 TiB
- 32 KiB — 128 TiB
- 64 KiB — 258 TiB

EBS Service Limitations

- EBS currently supports a maximum volume size of **64 TiB**.
 - You can create volumes larger than 64 TiB but the OS determines if it can read the entire capacity.
- Boot volumes must use either the MBR or GPT partitioning scheme
 - With MBR, boot volumes are limited to 2 TiB in size.
 - With GPT, boot volumes can be up to 64 TiB in size when used with GRUB2 (Linux) or UEFI boot mode (Windows).
- Non-boot volumes that are 2 TiB (2048 GiB) or larger must use a GPT partition table to access the entire volume.
- EBS-imposed limit on volume size (64 TiB) is currently equal to the maximum size enabled by 16-KiB data blocks.

Amazon Data Lifecycle Manager



Amazon Data Lifecycle Manager automates the creation, retention, and deletion of EBS snapshots and EBS-backed AMIs

Policies to define your backup creation and retention requirements and you have two options:

- **Default Policies** (back up all volumes and instances in a Region that do not have recent backups)
 - Default policy for EBS snapshots
 - Default policy for EBS-backed AMIs
- **Custom Policies**
 - EBS snapshot policy
 - EBS-backed AMI policy
 - Cross-account copy event policy

```
aws dlm create-lifecycle-policy \  
--description "My first policy" \  
--state ENABLED \  
--execution-role-arn arn:aws:iam::12345678910:role/AWSDataLifecycleManagerDefaultRole \  
--policy-details '  
{  
  "ResourceTypes": ["VOLUME"],  
  "TargetTags": [  
    {"Key": "costCenter", "Value": "115"}  
  ],  
  "Schedules": [{  
    "Name": "DailySnapshots",  
    "CopyTags": true,  
    "TagsToAdd": [  
      {  
        "Key": "type",  
        "Value": "myDailySnapshot"  
      }  
    ],  
    "CreateRule": {  
      "Interval": 24,  
      "IntervalUnit": "HOURS",  
      "Times": ["03:00"]  
    },  
    "RetainRule": {"Count": 5}  
  }  
}]  
'
```